

COM S 474/574: Introduction to Machine Learning

Project Proposal

1. Project Title

Remaining Useful Life (RUL) Prediction for Jet Engines using NASA CMAPSS Dataset

2. Team Members

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3. Project Details

3.1 Project Objective

The objective of this project is to build a machine learning model that predicts the Remaining Useful Life (RUL) of a jet engine using multivariate sensor time-series data.

- **Problem:** Unplanned engine failures are costly and unsafe. Maintenance teams require early predictions to schedule repairs before failure.
- **Importance:** Predictive maintenance reduces downtime, lowers cost, and improves safety by enabling condition-based maintenance instead of fixed schedules.
- **Why Machine Learning:** Engine degradation depends on complex temporal patterns across multiple sensors. Machine learning models can learn these patterns from historical run-to-failure data and generalize to new engines.

3.2 Dataset

We will use the NASA CMAPSS turbofan engine degradation dataset, a benchmark dataset for predictive maintenance research.

- The dataset contains multiple engines run until failure.
- Each cycle includes an engine ID, operational settings, and 21 sensor measurements.
- The label is Remaining Useful Life (RUL), computed as the difference between the final failure cycle and the current cycle.
- Data will be split by engine ID (not random rows) into training, validation, and testing sets to prevent data leakage.

3.3 Machine Learning Algorithm

This is a time-series regression problem.

- We will use recent sequences of sensor readings as input features to capture how engine behavior changes over time.
- The primary model will be a 1D Convolutional Neural Network (CNN) to predict Remaining Useful Life.
- Baseline models such as linear regression and random forest regression will be implemented for comparison.

Sequence models are appropriate because engine health depends on how sensor values evolve over time rather than a single observation.

3.4 Expected Outcomes

We expect to:

- Build a complete machine learning pipeline for RUL prediction.
- Train a deep learning model that outperforms baseline models.
- Evaluate performance using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
- Provide error analysis comparing predicted and true RUL values.